

Problem D. Vlad and the Best of Five

Time Limit 1000 ms

Mem Limit 262144 kB

Vladislav has a string of length 5, whose characters are each either A or B.

Which letter appears most frequently: A or B?

Input

The first line of the input contains an integer t ($1 \leq t \leq 32$) — the number of test cases.

The only line of each test case contains a string of length 5 consisting of letters A and B.

All t strings in a test are different (distinct).

Output

For each test case, output one letter (A or B) denoting the character that appears most frequently in the string.

Examples

Input	Output
8 ABABB ABABA BBBAB AAAAA BBBBB BABAA AAAAB BAAAA	B A B A B A A A